

Predicting nitrous oxide (N₂O) emissions from agriculture

Andreas Hiropedi¹, Andrew Ding¹, Fabio Turco², Lorenz Allemann², Valentina Boeva³
¹ETH Zurich Student, D-INFK ²ETH Zurich, D-USYS ³ETH AI Center, ETH Zurich

1. Problem Statement

Nitrous oxide (N₂O) is the 3rd most important greenhouse gas (GHG) and most of its anthropogenic emissions are attributed to agriculture. Due to the large temporal and spatial variability of N₂O emissions as well as data scarcity from field studies, our ability to predict N₂O is still poor, hindering our capacity to estimate overall losses and identify potential mitigation strategies.

Main goal: develop robust approaches to predict N₂O emissions from agricultural ecosystems.

2. Related work

- Saha et al. (2021) used **Random Forest** to predict N₂O emissions, capturing complex nonlinear relationships and achieving R² of **0.38**.
- Gnisia et al. (2025) found that simpler models (**Random Forest**, **XGBoost**) generally outperform deep learning models for frequent flux prediction such as weekly fluxes, while deep learning approaches such as **feed-forward neural networks** excel at predicting annual fluxes.

3. Data

The data we used for our project is a mixture of published open-source data and not yet published data collected by the Grassland sciences group at ETH in the last decade at different agricultural sites in Switzerland. We explored a total of 6 datasets which were either **cropland**, **grassland** or a mix of the two (**cropland with a temporary period of grassland**). Each dataset consists of half-hourly N₂O emissions, weather data, and agronomic management data from multiple sites and years.

4. Methods

The data were preprocessed following a consistent workflow across sites. The base datasets were first preprocessed to generate lagged meteorological and flux features (*NEE, GPP, RECO, radiation, temperature, VPD, soil moisture/temperature at multiple depths, precipitation*) at 1, 3, 5, and 7-day offsets, along with 3/5/7-day rolling means and sums. The temporal nature of field management variables were encoded as days-since events (*mowing, organic/mineral fertilization; capped at 30 days*). Fertilizer N application (kg N/ha) was merged from external logs, and exponential decay terms with 3, 7, and 14-day half-lives were derived. The target N₂O flux was log-transformed, with non-positive values set to zero. Daily datasets were produced by aggregating the base preprocessed data into daily means (continuous variables) and daily sums (precipitation). Models were then trained using Random Forest and XGBoost under three feature sets: **base** (*no lags*), **lag** (*base + lag variables omitting rolling lags*), and **augmented lag** (*base + all lag features*).

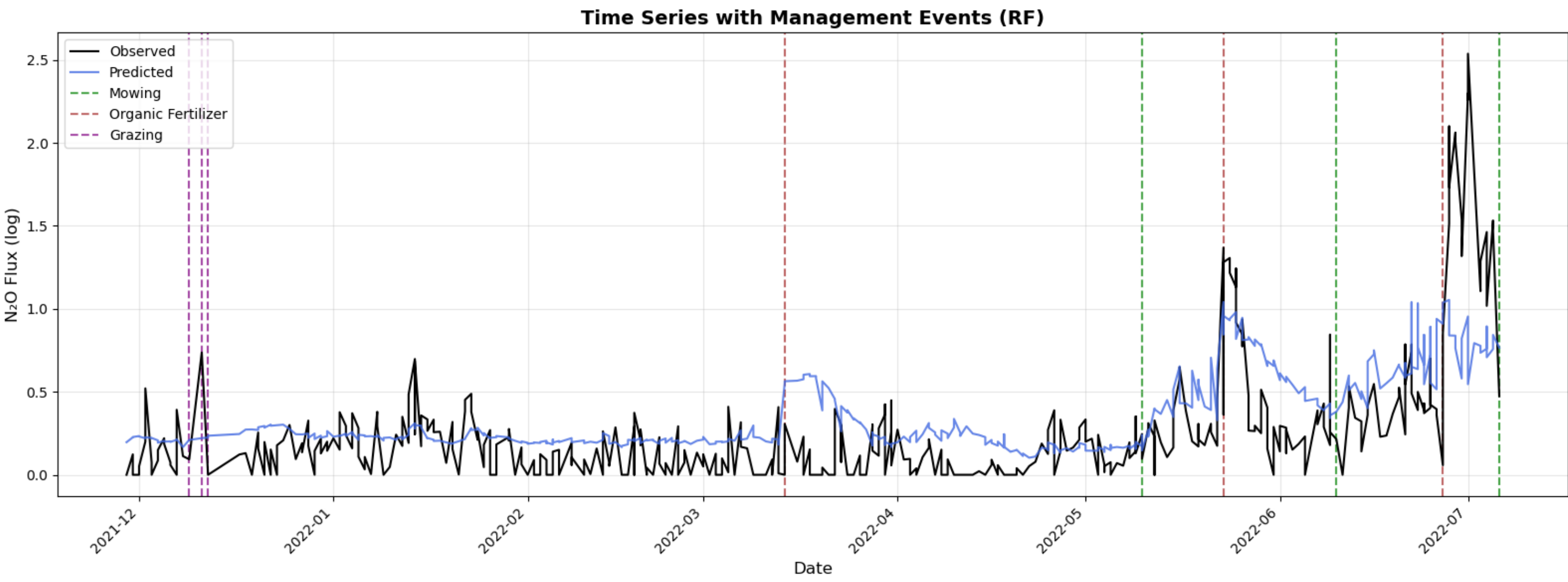


Figure 3. Test-set Random Forest predictions for Chamau with management events overlayed.

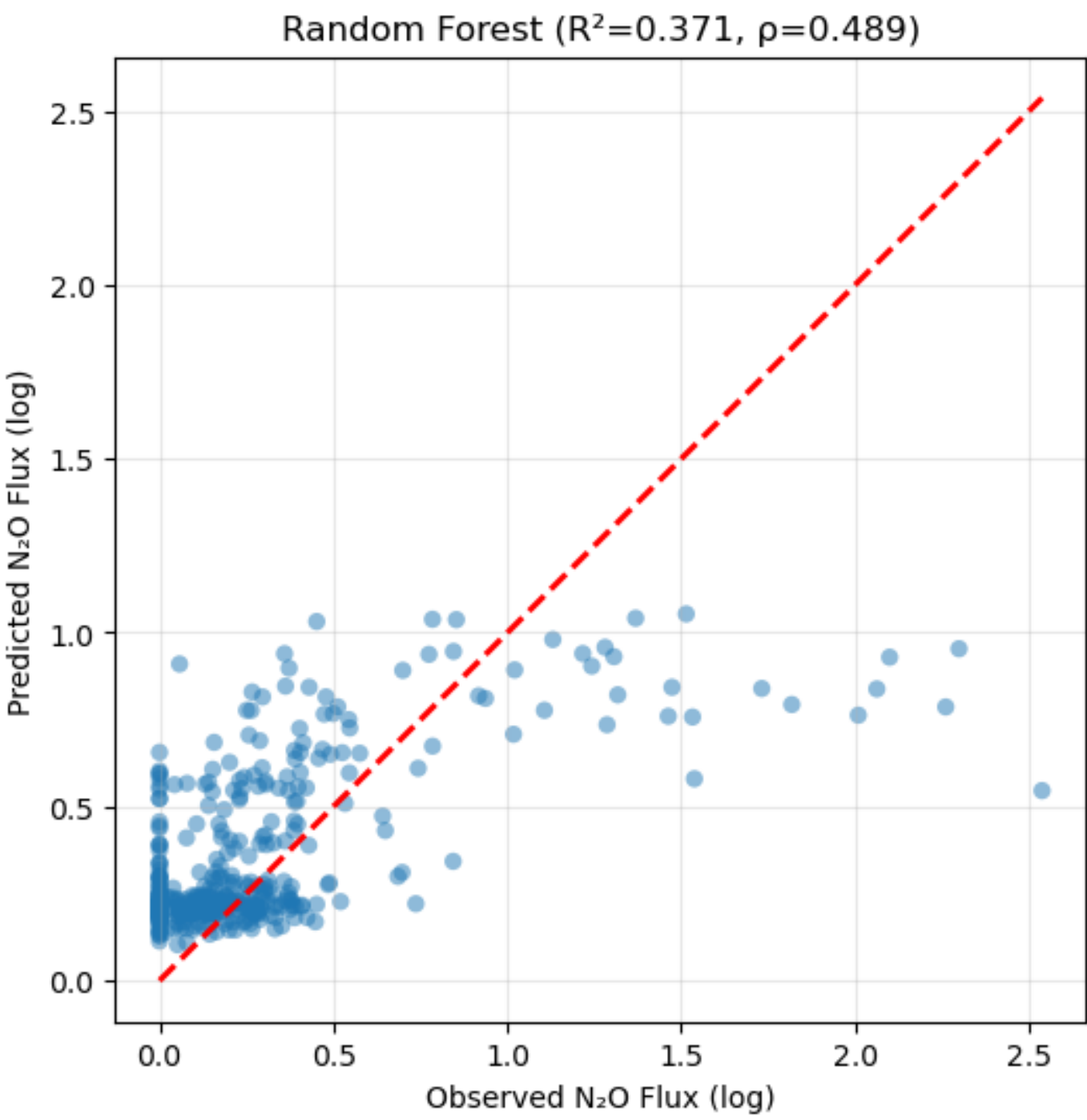


Figure 1: Scatterplot of RF model accuracy for Chamau

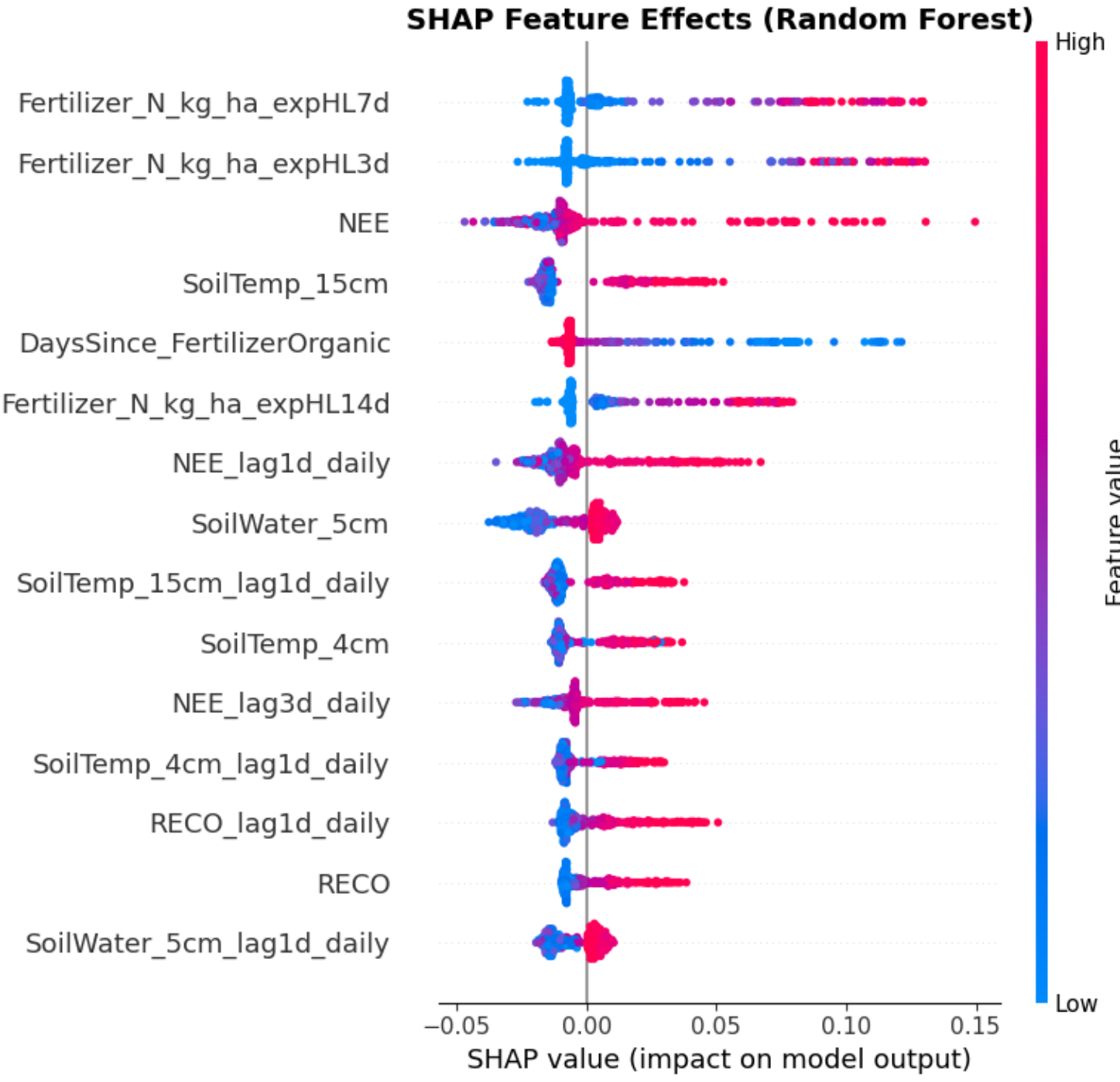


Figure 2: SHAP values for the best result of our best performing model.

5. Results

We provide a brief summary of our results for the Chamau dataset, as it has the largest sample size. Within the Chamau dataset, the daily-lag configuration achieved the strongest performance.

Model	Accuracy (R ²)
ElasticNet	0.07
Random Forest	0.37
XGBoost	0.34
Multi-layer Perceptron	0.22
Support Vector Machine	0.33

Table 1: Best observed results for the Chamau dataset.

The SHAP summary indicates that decayed N fertilizer presence is the strongest driver of predicted N₂O emissions, followed by NEE and soil temperature/water conditions. The top predictors are also consistent with what we found in other model settings, which suggests that they play a key role in predicting and managing N₂O emissions.

6. Conclusions

After fitting the models, we found that the daily aggregated datasets with the lag setting provided the strongest results. We theorize that by aggregating by day, we reduce the amount of noise which lead to more accurate predictions. Nevertheless, it remains difficult to accurately predict N₂O emissions, due to high variability in environmental drivers. Overall, fertilizer availability, NEE, and soil temperature emerged as the strongest predictors of emissions.

7. Future Work

- One direction for further research could involve finetuning models using only the most significant features extracted using our SHAP values analysis
- Another direction could be to combine the datasets (and therefore amass more data) and then test the existing models on these combined datasets

References

Gregor Gnisia, Jan Weik, Reiner Ruser, Lisa Essich, Iris Lewandowski, and Anthony Stein. Machine learning-based prediction of nitrous oxide emissions from arable farming: Exploring management practices as predictor variables. *Ecological Indicators*, 172:113233, 2025.

Debasish Saha, Bruno Basso, and G Philip Robertson. Machine learning improves predictions of agricultural nitrous oxide (n2o) emissions from intensively managed cropping systems. *Environmental Research Letters*, 16(2):024004, 2021.

Contributions

AH, AD: Methodology, Software, Visualization, Investigation, Writing
FT, LA: Conceptualization, Supervision
VB: Supervision, Methodology, Project Administration